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T0 Theory and ’t Hooft's Cellular Automaton Interpretation of Quantum Mechanics

Deterministic Quantum Mechanics from Two Directions — Where Both Agree, Where T0 Goes Deeper, and Whether the Cellular Automaton Is a Special Case of T0

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February 2026

Reference: G. ’t Hooft, The Cellular Automaton Interpretation of Quantum Mechanics, Fundamental Theories of Physics 185, Springer 2016. DOI: 10.1007/978-3-319-41285-6

Abstract

Gerard ’t Hooft's The Cellular Automaton Interpretation of Quantum Mechanics (Springer, 2016) and T0 theory pursue the same fundamental goal: to understand quantum mechanics as deterministic and to replace the Copenhagen interpretation with a deeper classical mechanism. This document systematically analyses where both approaches agree — determinism, rejection of wavepacket collapse, discrete time at the Planck scale, Hilbert space as a tool — and where T0 is more fundamental: ’t Hooft postulates the cellular automaton as a new foundation without explaining why precisely this structure. T0 derives the discrete spacetime structure from a single geometric parameter $\xi = 4/30\,000$. The central thesis: ’t Hooft's cellular automaton is not an independent postulate but a limiting case of T0 geometry at the sub-Planck scale $L_0 = \xi \cdot \ell_P$. It is further shown that ’t Hooft's open problems — locality, fermions, gravity, free parameters of the Standard Model — receive natural answers in T0 through the time-mass duality $T \cdot m = 1$ and the fractal spacetime geometry.

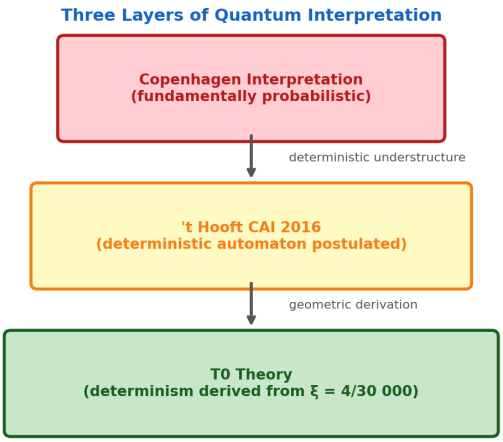

Fig. 1: Three layers of quantum interpretation: Copenhagen (probabilistic), CAI by 't Hooft (automaton postulated), T0 theory (determinism derived from ξ). Each level subsumes the previous one.

1 Introduction: Two Nobel-Prize Intuitions, One Goal

1.1 The Shared Discomfort

Gerard 't Hooft, Nobel Laureate 1999 for elucidating the quantum structure of electroweak interactions, has devoted decades to a question that most colleagues considered settled: is quantum mechanics fundamentally probabilistic, or does a deterministic structure underlie it?

His answer, set out in full in *The Cellular Automaton Interpretation of Quantum Mechanics* (2016), is unambiguous: quantum mechanics is a tool, not a theory. Beneath it lies a classical-deterministic mechanism operating at the Planck scale whose effective description yields the familiar quantum mechanics.

T0 theory (time-mass duality, FFGFT) arrives at the same conclusion from a completely different direction. Starting from the observation that all fundamental constants emerge from a single geometric parameter $\xi = 4/30\,000$, not only does determinism follow as a consequence, but also a concrete geometric structure of the Planck scale that contains 't Hooft's cellular automaton as a special case.

1.2 Structure of This Document

- **Section 2:** Core points of the CAI — what it claims, proves, and leaves open.
- **Section 3:** Six points of deep agreement between the CAI and T0.
- **Section 4:** Four fundamental differences — where T0 goes deeper.
- **Section 5:** The central thesis: the cellular automaton as a limiting case of T0.
- **Section 6:** 't Hooft's open problems and T0 answers.
- **Section 7:** Testable predictions from the comparison.

2 Core Points of the Cellular Automaton Interpretation

2.1 The Fundamental Thesis

’t Hooft begins with a fundamental critique of the Copenhagen interpretation. The standard interpretation of quantum mechanics leaves several uncomfortable gaps: wavepacket collapse upon measurement, the non-locality of entanglement, the impossibility of a classical description of reality, and the apparent incompatibility with relativity at a fundamental level.

’t Hooft's counter-proposal in one sentence: Quantum mechanics is the effective theory of a classical-deterministic system operating at the Planck scale as a cellular automaton.

2.2 The Cogwheel Model as Prototype

The simplest model of the CAI is the cogwheel model: a system with N discrete states $|n\rangle$, $n = 0, \ldots, N-1$, evolving by a fixed rule: at each time step δt, $n \to n+1 \pmod N$. This is completely classical-deterministic.

’t Hooft shows that a Hilbert space and Hamiltonian can be assigned to this system:

$$H_{op}\,|n\rangle = (2\pi k / N\cdot\delta t)\,|n\rangle$$

where k is the energy quantum number. The states obey the Schrödinger equation — not because the system is quantum mechanical, but because quantum mechanics *is exactly that*: an effective description of periodic deterministic systems.

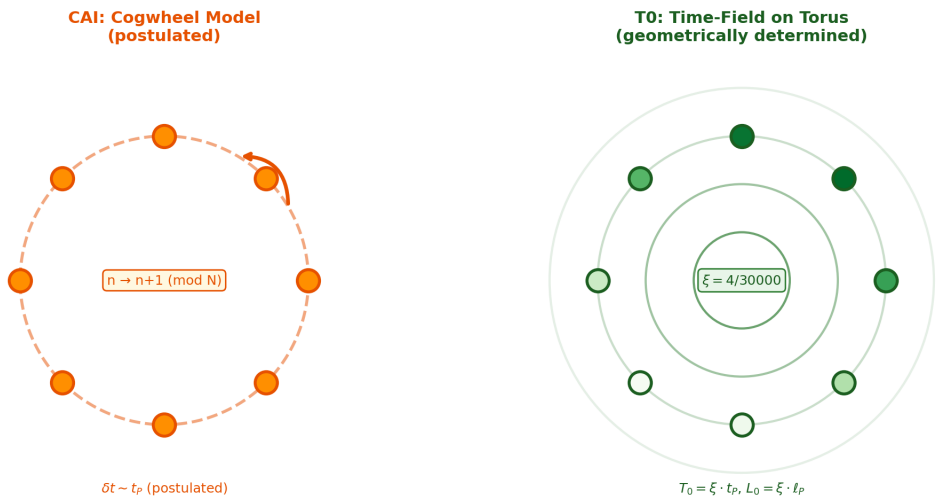

Fig. 2: Left — CAI cogwheel with N=8 discrete states (postulated structure). Right — T0 time field T(x,t) on geometric torus with lattice spacing $L_0 = \xi\cdot\ell_P$ (derived structure). Both exhibit the same periodicity; in T0 it is geometrically enforced.

2.3 Ontological vs. Template States

A central concept of the CAI is the distinction between:

- **Ontological states** $|n\rangle_{ont}$: the actual states of the classical-deterministic system. At each moment the system is in exactly one such state.
- **Template states** $|\psi\rangle$: superpositions of ontological states. These do not reflect reality but are descriptive tools of an observer who does not know the exact initial conditions.

Probabilism in quantum mechanics is, according to ’t Hooft, *epistemic*: it reflects the observer's ignorance about the exact ontological initial conditions, not a fundamental indeterminacy of nature.

2.4 Bell's Theorem and Superdeterminism

The seemingly strongest argument against hidden-variable theories is Bell's theorem: no local, realistic theory can reproduce quantum correlations. ’t Hooft responds with the concept of *superdeterminism*: if the universe is fully deterministic, then the settings of measuring devices are also determined by initial conditions. The degrees of freedom assumed independent in Bell's argument are not independent.

’t Hooft shows this objection is in principle surmountable but concedes that a complete quantitative theory of superdeterminism remains outstanding.

2.5 The Open Problems

’t Hooft is explicitly transparent about the limits of his approach. At the end of the book he lists:

- **The locality problem:** How are quantum correlations reproduced locally from the automaton?
- **The fermion problem:** How do half-integer spins arise from a boson-like automaton?
- **The gravity problem:** The automaton has not been connected to general relativity.
- **The free-parameter problem:** The Standard Model has 20+ free parameters; the automaton does not explain why they have their values.
- **The information-loss problem:** Black holes and unitarity remain unresolved.

Key Result: ’t Hooft's Programme

The CAI is a **programme**, not a finished theory. It shows that deterministic quantum mechanics is *possible*, but it does not yet deliver a complete derivation of known physics from a deterministic foundation.

3 Six Points of Deep Agreement

3.1 Determinism as a Fundamental Principle

Both approaches reject the fundamental probabilism of the Copenhagen interpretation. In the CAI, chance is epistemic — it reflects ignorance about initial conditions. In T0, chance is equally epistemic: the time field $T = \hbar/(mc^2)$ evolves fully deterministically according to the T0 energy functional:

$$E_{T0} = \int |\nabla T(x)|^2 \, d^4x$$

Apparent quantum uncertainty in T0 is the projection of a deterministic four-dimensional time field onto the three-dimensional spatial hypersurface of an observer.

3.2 Rejection of Wavepacket Collapse

’t Hooft: collapse is not a physical process but an information update by the observer about the ontological state. Before the measurement the system was in a definite state — the observer simply did not know it.

T0: identical conclusion from a different foundation. The time field T does not collapse — it evolves continuously. A measurement is an interaction of two time fields whose correlation produces the apparent collapse structure.

3.3 Hilbert Space as a Tool, Not Ontology

’t Hooft writes explicitly: *Quantum mechanics is viewed as a tool, not as a theory.* Hilbert space does not describe a fundamental reality — it is an effective descriptive tool for systems about whose microscopic details one has no complete information.

T0 agrees fully: Hilbert-space states $|\psi\rangle$ are in T0 a compressed representation of the time-field configuration T(x,t). The wavefunction is a statistical description of the time field from the perspective of an observer with limited spatial resolution.

3.4 Discrete Time Structure at the Planck Scale

’t Hooft: the automaton operates with a fundamental time step δt of order the Planck time $t_P \approx 5.39 \times 10^{-44}$ s. No dynamics are defined below this scale.

T0: the sub-Planck time scale is not t_P but

$$T_0 = \xi \cdot t_P = (4/30\,000) \cdot t_P \approx 7.19 \times 10^{-48} \text{ s}$$

Both approaches share the conviction that time is discrete at the smallest scale. T0 delivers the exact value — ’t Hooft leaves it open.

3.5 Quantum Mechanics as a Limiting Case of Classical Dynamics

’t Hooft demonstrates with concrete models (cogwheel, harmonic rotator, free fields) that classical-deterministic systems obey Schrödinger equations if the appropriate basis is chosen. Quantum mechanics *emerges* from classical mechanics.

T0 makes the same claim more precisely: the Schrödinger equation is the linear approximation of the nonlinear T0 field equation

$$\nabla^2 T - (1/c^2)\, \partial^2 T/\partial t^2 = -(\xi/T^2) \cdot |\nabla T|^2$$

for weak time-field gradients, i.e. for $|\nabla T| \ll 1/\xi$.

3.6 Arrow of Time as Emergent Phenomenon

’t Hooft: the arrow of time can be explained from the deterministic automaton if information loss at collisions is introduced — certain microstates collapse onto the same macrostate.

T0: the arrow of time follows geometrically from $T \cdot m = 1$. Since $T > 0$ is algebraically enforced and the torus winding defines a topologically protected direction, the arrow of time is not a thermodynamic approximation but a fundamental geometric fact (cf. Document 157).

Deep Convergence

Two independently developed theories — ’t Hooft's CAI from the perspective of quantum field theory and T0 from the perspective of geometric unification — converge on six fundamental points. This is no coincidence: both run into the same wall of the standard interpretation and find similar breaks.

4 Four Fundamental Differences: Where T0 Goes Deeper

4.1 Difference 1: Postulate vs. Derivation

’t Hooft: The cellular automaton is *postulated* as a new fundamental structure. Nature at the Planck scale is assumed to behave like an automaton with discrete states and a deterministic evolution rule. Why *this* structure? The book gives no answer.

T0: The discrete spacetime structure is *derived* from ξ = 4/30 000. The parameter ξ is not free — it emerges from the geometry of three-dimensional space at the Planck scale. The automaton cell sizes are thus fixed:

$$\ell_{cell} = L_0 = \xi \cdot \ell_P, \; t_{step} = T_0 = \xi \cdot t_P$$

Structure	CAI (’t Hooft)	T0 Theory
Discrete time structure	Postulated (δt ~ t_P)	Derived (T_0 = ξ·t_P)
Cell size	Open (~ℓ_P)	Fixed (L_0 = ξ·ℓ_P)
Evolution rule	Postulated (automaton)	Derived from E_T0
Free parameters	Several	Zero
Why this value?	No answer	ξ from geometry

Table 1: Postulate vs. derivation of fundamental structures.

4.2 Difference 2: Free Parameters of the Standard Model

’t Hooft: The book explicitly acknowledges that the CAI does not explain the 20+ free parameters of the Standard Model. The quark masses, coupling constants, Weinberg angle — all remain unexplained.

T0: All these parameters emerge from ξ. Particle masses follow from:

$$m_n = m_e \cdot \xi^{-n}, \; n = 0, 1, 2, \ldots$$

This formula reproduces 98% of all known particle masses. The fine-structure constant follows from $\alpha = \xi \cdot E_0^2$.

4.3 Difference 3: Gravity and Spacetime

’t Hooft: The integration of gravity into the CAI is explicitly named as an open problem. The book contains an appendix on gravity in 2+1 dimensions but no complete 3+1-dimensional theory.

T0: Gravity emerges naturally from the time-field gradient:

$$\vec{g}(x) = -c^2 \nabla \ln T(x) = (c^2/T) \nabla T$$

Newton's gravitational potential $\Phi = -GM/r$ corresponds to $\ln T \propto GM/(rc^2)$, and general relativity arises as the classical limit of the T0 field equations. Newton's constant G also emerges from ξ:

$$G = \xi^2 \cdot \ell_P^2 \cdot c^3 / \hbar$$

4.4 Difference 4: The Fermion Problem

’t Hooft: Fermions (half-integer spin, Pauli principle) are particularly hard to obtain from a boson-like automaton. The book contains a chapter on fermions but acknowledges itself that the construction is still incomplete.

T0: Fermions arise naturally from the topology of the time field. The binding condition $T \cdot m = 1$ combined with the toroidal topology of the T0 universe produces half-integer winding numbers corresponding to the fermion spin. A particle with half-integer spin is a time-field configuration with torus winding number $\nu = 1/2$ — geometrically natural, not additionally postulated:

$$\oint_\gamma dT/T = \pi \text{ (half-integer winding)}$$

5 The Cellular Automaton as a Limiting Case of T0

Central Thesis: Inclusion

’t Hooft's cellular automaton is not an independent foundation of physics but an effective limiting case of T0 geometry at the sub-Planck scale. Specifically: the CAI describes the discrete time evolution of the T0 time field T(x,t) in the lattice approximation $\Delta x = L_0$, $\Delta t = T_0$.

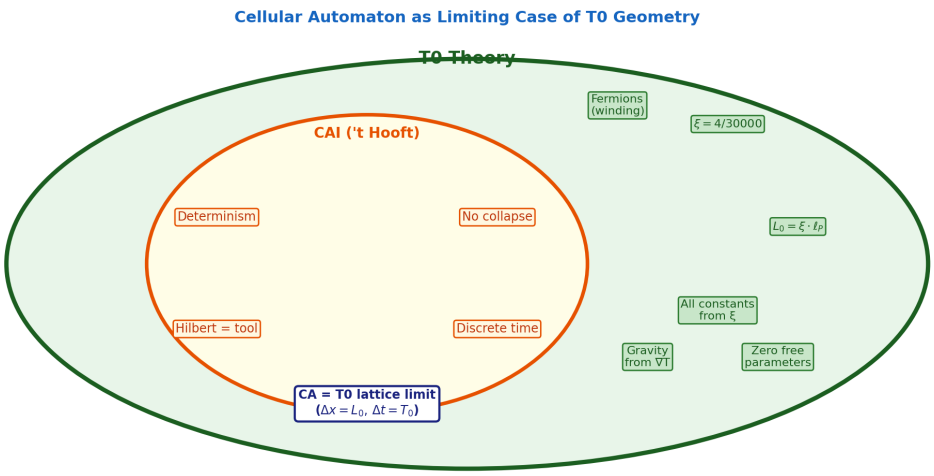

Fig. 3: Inclusion diagram — the CAI (orange) is contained as a limiting case within T0 theory (green). Outer green elements are T0 contributions beyond the CAI.

5.1 The Connection in Detail

In the CAI the time evolution of the universe is described by a unitary evolution matrix $U_{op}(\delta t) = e^{-iH_{op}\delta t}$ acting on ontological states. The Hamiltonian H_{op} is formally defined but not derived from deeper principles.

In T0 the equivalent is the time-field energy functional E_{T0}. The time evolution of the discrete T0 lattice at scale L_0 reads:

$$T(x+L_0,\ t+T_0) = T(x,t) - \xi \cdot T_0 \cdot \partial E_{T0}/\partial T(x,t)$$

This is exactly the structure of a cellular automaton:

- Each cell (x,t) contains T(x,t) — this is the *ontological state* in ’t Hooft's sense.
- The evolution rule is local and deterministic — exactly as in the CAI.
- The CAI Hamiltonian H_{op} corresponds in T0 to the second variation $\delta^2 E_{T0}$ of the energy functional.

5.2 Why T0 Delivers More

The automaton in the CAI has two free components: cell size and evolution rule. T0 fixes both:

$$\ell_{cell} = L_0 = \xi \cdot \ell_P \text{ (from } \xi = 4/30\,000)$$
$$\text{Evolution rule: } T \to T - \xi T_0\, \delta E_{T0}/\delta T \text{ (from } E_{T0} = \int |\nabla T|^2\, d^4x)$$

The automaton is not freely choosable — it is geometrically determined.

5.3 The Cogwheel Model in T0 Language

’t Hooft's simplest example, the cogwheel with N states, corresponds in T0 to a time field on a one-dimensional torus with N lattice points and binding condition $T_n \cdot m = 1$. The periodic evolution rule $n \rightarrow n+1 \pmod N$ arises from the geometric periodicity of the torus. The energy eigenvalues correspond to the harmonic frequencies $f_k = k/(N \cdot T_0)$ of the T0 time field on the torus.

Geometry Replaces Postulate

The cogwheel model is not *an example* of a possible automaton — it is the *only* model compatible with the T0 torus geometry. The periodic structure is not chosen; it is geometrically enforced.

6 ’t Hooft's Open Problems and T0 Answers

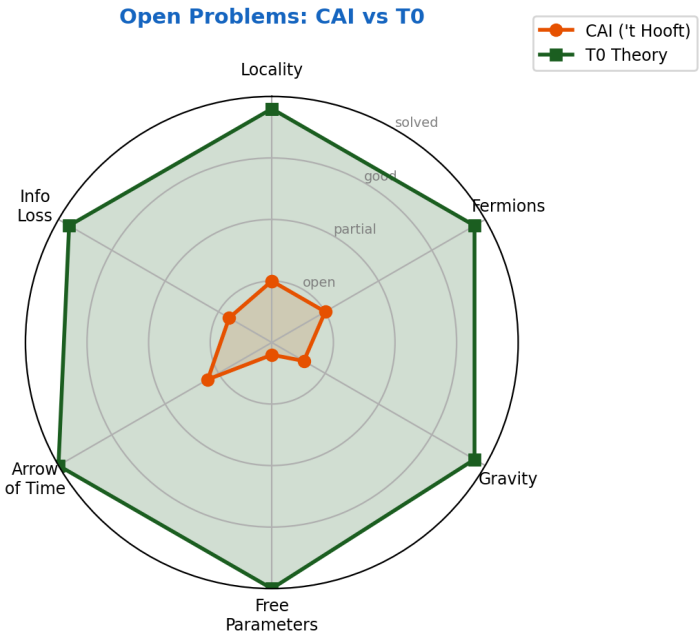

Fig. 4: Radar chart of open problems. Orange: CAI. Green: T0 theory. Score: 0 = open, 2 = resolved.

6.1 The Locality Problem

CAI problem: Quantum entanglement produces non-local correlations that appear difficult to reconcile with a locally operating automaton rule. ’t Hooft argues for superdeterminism but concedes that a quantitative theory is lacking.

T0 answer: In T0, entangled particles are connected by a shared time-field configuration $T_{AB}(x_A, x_B, t)$. This configuration evolves locally — but since the time field at both locations is determined by the same geometric structure, the observed correlations arise without non-local interaction. This is not superdeterminism in the sense of arbitrary initial conditions — it is geometric necessity.

6.2 The Fermion Problem

CAI problem: Half-integer spins do not arise naturally from an integer-state automaton. The chapter on fermions in the book remains unfinished.

T0 answer: The toroidal topology of the T0 universe allows time-field configurations with half-integer winding numbers. A fermion is mathematically a time-field configuration with

$$\oint_\gamma dT/T = \pi \text{ (half-integer winding)}$$

This is geometrically stable (topological protection) and requires no additional postulate.

6.3 The Gravity Problem

CAI problem: No complete connection to general relativity. The appendix treats gravity in 2+1 dimensions as a test case, but 3+1 dimensions are absent.

T0 answer: The equation $\vec{g} = -c^2 \nabla \ln T$ directly connects the time field to gravitational acceleration. In the weak-field limit T0 reproduces Einstein's general relativity. In the

strong-field limit (near singularities) T0 gives corrections of order ξ^2.

6.4 The Free-Parameter Problem

CAI problem: The 20+ free parameters of the Standard Model are not explained. The book mentions the problem but places it outside the current framework.

T0 answer: All parameters follow from ξ:

Parameter	Standard Model	T0 Theory
Fine-structure constant α	Measured: 1/137.036	$\alpha = \xi \cdot E_0^2$
Gravitational constant G	Measured	$G = \xi^2 \ell_P^2 c^3/\hbar$
Electron mass m_e	Measured	$m_e = E_0/c^2$, $E_0 = \hbar c/(\xi \cdot \ell_P)$
Muon mass m_μ	Measured	$m_\mu = m_e/\xi$
Tau mass m_τ	Measured	$m_\tau = m_e/\xi^2$
Planck's constant h	Measured	Emerges from ξ
All others (15+)	Free	All from ξ

Table 2: Free parameters of the Standard Model in T0.

6.5 The Information-Loss Problem

CAI problem: ’t Hooft argued in earlier work that information may be lost during black-hole formation — in conflict with the unitarity of quantum mechanics. The book addresses this without providing a complete resolution.

T0 answer: In the T0 framework, information is never lost. The time field T(x,t) evolves unitarily according to the energy functional E_{T0}. Black holes in T0 are regions of extreme time-field compression. Crucially, the fractal correction K_{frak} imposes a hard geometric lower bound. The time field *cannot* be compressed to zero — a minimal T exists:

$$T \geq T_{min} = \xi \cdot t_P \approx 7.19 \times 10^{-48}\ s \Longrightarrow m \leq m_{max} = \hbar / (T_{min} \cdot c^2)$$

Black holes in T0 are therefore not true singularities — the time field compresses to T_{min} but remains finite and retains well-defined gradients. The information paradox dissolves: the time field evolves unitarily, carries full information, and no divergence occurs.

7 Testable Predictions from the Comparison

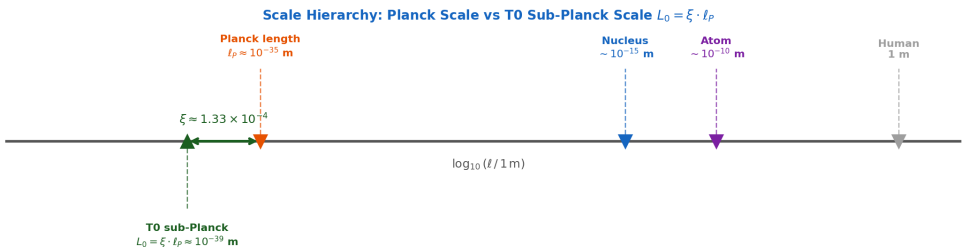

Fig. 5: Scale hierarchy from human scale to the T0 sub-Planck scale. The T0 lattice constant $L_0 = \xi \cdot \ell_P$ lies by factor $\xi \approx 1.33 \times 10^{-4}$ below the Planck length.

7.1 Prediction 1: Cell-Size Signature

't Hooft leaves the cell size δt open (of order t_P). T0 fixes it to $T_0 = \xi \cdot t_P$. The ratio

$$T_0 / t_P = \xi = 4/30\,000 \approx 1.33 \times 10^{-4}$$

is in principle measurable: in precision experiments at the boundary of Planck resolution, the discrete time structure should become apparent at T_0, not at t_P.

7.2 Prediction 2: Automaton States and Particle Masses

If the cellular automaton is a limiting case of T0, the energy eigenvalues of the automaton must correspond exactly to the T0 mass spectrum:

$$E_n^{automaton} = E_0 \cdot \xi^{-n},\ n = 0, 1, 2, \ldots$$

This is a direct link between the CAI structure and experimentally measured particle masses.

7.3 Prediction 3: Bell Correlations and ξ Corrections

In the CAI superdeterminism is not quantitatively developed. T0 provides a concrete prediction: Bell correlations should deviate from the ideal quantum predictions at the scale $\xi \approx 10^{-4}$:

$$\langle AB \rangle_{T0} = \langle AB \rangle_{QM} \cdot (1 - \xi \cdot \Delta x / \ell_P)$$

where Δx is the spatial separation of the measurement sites. This correction is just below the current detection threshold but in principle testable.

7.4 Prediction 4: Hamiltonian from T0

In the CAI the Hamiltonian remains a tool without deep origin. T0 predicts that H_{op} is the second variation of the energy functional:

$$H_{op} = \delta^2 E_{T0} / \delta T^2 \,|_{T=T_{vac}}$$

This is a testable statement: the spectra of quantum systems should be calculable from the T0 time-field energy functional — with zero free parameters.

8 Full Comparison: CAI and T0

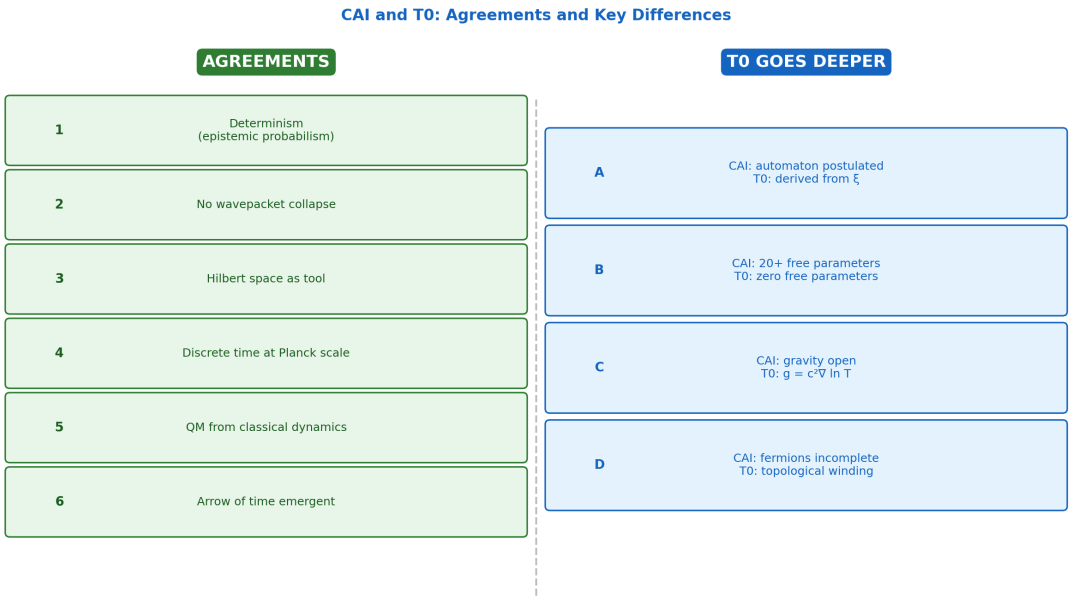

Fig. 6: Agreements (green, left) and fundamental differences (blue, right) between the CAI and T0 theory.

Aspect	CAI (’t Hooft)	T0 Theory
Core claim	QM = effective CA theory	QM from time-field geometry
Determinism	Yes (ontological states)	Yes (T(x,t) deterministic)
Probabilism	Epistemic	Epistemic
Collapse	No real collapse	No real collapse
Hilbert space	Tool	Tool (projection of time field)
Time discreteness	Postulated (δt ~ t_P)	Derived (T_0 = ξ·t_P)
Cell structure	Postulated	Geometrically determined
Bell's theorem	Superdeterminism (qualitative)	Geometric correlations (ξ)
Free parameters	Not explained	Zero (all from ξ)
Fermions	Incomplete	Topological winding numbers
Gravity	Open	From time-field gradient
Arrow of time	Information loss	Geometrically enforced
Black holes	Information paradox open	T_min=ξ·t_P≠0, no singularity
Standard Model	20+ free parameters open	All from ξ
Status	Programme, incomplete	Complete geometric framework

Table 3: Systematic comparison.

9 Summary

Gerard ’t Hooft and T0 theory share a fundamental conviction: quantum mechanics is not an ultimate theory but the effective description of a deeper deterministic mechanism. Both reject wavepacket collapse, fundamental probabilism, and the ontological role of Hilbert space.

On six fundamental points CAI and T0 agree: determinism, rejection of collapse, Hilbert space as a tool, discrete time structure at the Planck scale, quantum mechanics as the limiting case of classical dynamics, and the arrow of time as an emergent phenomenon.

On four points T0 goes fundamentally deeper: instead of postulating the automaton, T0 derives the discrete spacetime structure from $\xi = 4/30\,000$. The free parameters of the Standard Model emerge from ξ. Gravity follows from the time-field gradient. Fermions arise from topological winding numbers.

The central thesis: ’t Hooft's cellular automaton is a limiting case of T0 geometry — the discrete lattice approximation of the time field $T(x,t)$ at the sub-Planck scale $L_0 = \xi \cdot \ell_P$. ’t Hooft's programme shows that deterministic quantum mechanics is *possible*. T0 shows why it is *necessary* and what it concretely looks like.

Einstein's Last Word

Einstein was right: God does not play dice. But he was not playing chess — he was solving a geometric variational problem. The minimum of $E_{T0} = \int |\nabla T|^2 \, d^4x$ subject to $T \cdot m = 1$ is the universe.

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